Mathematical Modeling of Propeller Mass Imbalance

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The Centrifugal Forcing Function (Quadrature Relationship)

Propeller mass imbalance occurs when the center of mass of the rotating assembly is offset from the true axis of rotation due to manufacturing defects, material wear, or physical damage (e.g., a chipped blade).

This localized defect behaves as a point mass m_u located at a radial distance r from the rotational axis. As the rotor spins at an angular velocity $\Omega = 2\pi f_0$, it generates a continuous, rotating centrifugal force. The magnitude of this rotating vector scales with the square of the angular velocity ($F \propto \Omega^2$).

To accurately simulate the projection of this rotating force vector onto the stationary Cartesian reference frame of the UAV flight controller, the continuous forcing function must be modeled with a phase quadrature relationship between the radial axes. The time-domain forces applied to the structural X and Y axes are:

$$F_{imb,x}(t) = m_u r \Omega^2 \sin(\Omega t) \tag{1}$$

$$F_{imb,y}(t) = m_u r \Omega^2 \cos(\Omega t) \tag{2}$$

This 90° phase difference provides a distinct spatial signature that distinguishes a rotating unbalance from a unidirectional planar vibration. Because the unbalance produces no lift component, the axial force is strictly zero ($F_{imb,z} = 0$).

Structural LTI Convolution and Accelerance Scaling

These continuous forcing functions are passed through the Linear Time-Invariant (LTI) structural transfer function $H(\omega)$ of the UAV chassis. The output acceleration $a(t)$ is the convolution of the input force and the structure's impulse response $h(t)$.

It is critical to account for how the vibration amplitude scales with RPM. The physical force scales as Ω^2. However, the accelerance Frequency Response Function (FRF) $H(\omega)$ roughly scales as Ω^2 well below the chassis structural resonance, and flattens out above it. Consequently, the measured acceleration response roughly scales as Ω^4 below resonance and Ω^2 above resonance.

To prevent detection thresholds from failing as the motor sweeps through different RPM ranges, the signal processing pipeline utilizes Order Normalization, ensuring the $1\times$ fundamental peak is evaluated relative to the instantaneous running speed.

Frequency-Domain Noise and Processing Gain

The raw time-domain signal inherently includes parameterized Gaussian environmental and sensor noise $\eta(t)$ with standard deviation σ. When evaluating the signal in the frequency domain, the presence of random-phase noise dictates that the magnitudes cannot be linearly added. Instead, the total expected power of the frequency spectrum $A(\omega)$ is the sum of the structural response power and the noise variance floor $\Gamma(\omega)$:

$$E[|A(\omega)|^2] = |H(\omega) \cdot F_{imb}(\omega)|^2 + E[|\Gamma(\omega)|^2] \tag{3}$$

To systematically evaluate the performance limits of the fault detection algorithm under varying noise levels, the theoretical Signal-to-Noise Ratio (SNR) within a specific FFT bin must be established. For a purely sinusoidal unbalance signal of amplitude A submerged in white noise of standard deviation σ, an N-point Fast Fourier Transform (FFT) yields a processing gain of $N/2$. The expected bin SNR is:

$$SNR_{bin} = \left(\frac{A^2}{2\sigma^2}\right)\frac{N}{2} \tag{4}$$

This theoretical limit serves as the validation curve against which the empirical simulation outputs will be plotted to quantify the algorithm's robustness.

Residual Unbalance and Threshold Classification

To maintain rigorous mechanical realism, a "healthy" propeller cannot be modeled with zero unbalance ($m_u = 0$). Due to standard manufacturing tolerances, all commercial propellers exhibit a microscopic residual unbalance.

The simulation logic defines the baseline state by drawing a normal residual unbalance $m_{u,healthy}$ independently for each motor from a defined manufacturing tolerance distribution. A fault state is defined by a physically severe mass loss: $m_{u,fault} \gg m_{u,healthy}$.

Consequently, the fault detection algorithm does not simply trigger upon the presence of a 1Ω peak, as this frequency exists in all operational states. Instead, classification relies on dynamic ratio thresholding. A fault is only flagged when the magnitude of the 1Ω component significantly exceeds the established residual

baseline for that specific RPM order, effectively separating a damaged blade from the inherent physical imperfections of a healthy propulsion system.